SaayaShift

Heat-aware shift planning | Karachi-first | HackInsight 2026

WORKING BUILD - NOT FINAL SUBMISSION | 19 October 2026 | Team: TBD

For a supervisor scheduling outdoor field work, compare complete work windows within operating constraints using forecast apparent temperature. Lower modeled heat exposure is not a declaration of medical safety.

1.113 °C	**0.342 °C**	**26,304 hours**
Declared H-24 baseline MAE	Ridge model MAE	Three local years of weather

Evidence and reproducibility

Original Open-Meteo ICON Global historical forecasts (2023-2025), a live outlook, and 25,075 OSM mapped geometries were acquired during the event. Raw bytes, UTC fetch times, queries, licenses and SHA-256 hashes are retained. Raw, cleaned and processed layers feed DuckDB and Streamlit. All six data tables rebuilt byte-identically from verified raw cache, with no network calls. Nineteen automated checks passed; human acceptance is pending.

Scientific result, with limits

Fixed development in 2023-2024; untouched 2025 test with 8,760 hours. MAE improvement is 0.772 °C (paired 95% block-bootstrap interval 0.671-0.887 °C). Model MAE interval: 0.322-0.363 °C. The nominal 90% interval covered only 85.0%; winter was weakest (0.412 °C MAE; 78.2% coverage).

This is retrospective next-hour prediction of a stitched weather-model series, not station measurements, actual issue-time forecast validation, or worker-health outcomes. The planner uses provider forecasts separately. One Karachi grid cell does not establish worksite or nationwide validation.

Agents, operations and remaining gates

The quality agent uses Pandera and explicit fix/flag decisions; the first successful batch had no nonzero findings and no filled targets. The Insight prototype retrieves five permission-cleared NIOSH passages, cites exact sections and row IDs, calculates numbers in code, and abstains. Full language-model RAG remains unfinished. A real local Prefect smoke run completed; hourly host and the required 20-21 October history remain pending. Public Space, final tag, video and team handover are not complete.

Value, costs and next action

Potential payer: a municipal contractor, utility or NGO (unvalidated hypothesis). The research build incurred $0 in service charges; local energy, hardware and labor are unmeasured. Commercial Open-Meteo use requires different terms. Confirm an always-on host and eligible Space account, complete human checks, and freeze by 21 October 18:00 PKT; internal ship target is 21:00 PKT.

Reproduce: install requirements.txt in Python 3.12; run make data, make evaluate, make dashboard. Windows: .\make.cmd data (then evaluate/dashboard). Evidence and exact instructions are in the accompanying repository snapshot.

Sources: Open-Meteo (CC BY 4.0, noncommercial API); © OpenStreetMap contributors (ODbL); CDC/NIOSH 2016-106 §6.2.1 (public domain, no endorsement). Declaration commit: 4013d44, 19 Oct 10:23 PKT. Full URLs and original hashes: ATTRIBUTION.md and data/manifest.csv.